DCIFNet: Cross-Modal Fusion With Correction and Interaction for Optical–SAR Land Cover Classification

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***Abstract*—Land cover classification (LCC) based on remote sensing image segmentation is a prominent task of remote sensing data interpretation. The commonly used optical data are susceptible to the weather, so it has the potential to utilize complementary features from the supplementary synthetic aperture radar (SAR) data to enhance segmentation performance. However, current multimodal segmentation methods focus on the deep fusion of features, which usually ignores the significance of structural consistency information. In order to make use of the mutual correction and information exchange between multimodal data, we propose DCIFNet, a dual-stream correction-interaction-fusion multimodal LCC network. Specifically, we design a differential feature correction and enhancement module (DF-CEM) that leverages bidirectional differential features to correct multimodal features. In addition, for corrected feature pairs, we deploy a parallel attention interaction module (PAIM) to focus on the pixel-level feature correlation and achieve effective information exchange in both channel and spatial dimensions. Through the expert fusion module (EFM), DCIFNet leverages the gate network to attain a flexible and compact feature fusion between multimodal features. The experimental results show that our method achieves a superior performance compared with other multimodal fusion segmentation methods on three optical–SAR datasets. The source code of DCIFNet is publicly available at https://github.com/XD-MG/DCIFNet**

***Index Terms*—Land cover classification (LCC), multimodal fusion, optical, synthetic aperture radar (SAR).**

I. Introduction

LAND cover classification (LCC) is a fundamental task in remote sensing data interpretation, widely used in applications such as environmental monitoring and urban development planning. It involves the pixel-wise categorization of remote sensing imagery into predefined land cover classes, closely related to semantic segmentation in computer vision (CV). With the widespread adoption of convolutional neural networks (CNNs), numerous CNN-based segmentation frameworks have been proposed. Fully convolutional networks (FCNs) [1] marked a breakthrough by enabling end-to-end training for dense pixel-level prediction through the transformation of traditional classification networks into fully convolutional architectures. Subsequently, encoder–decoder architectures with skip connections [2] were introduced, significantly enhancing both spatial localization and contextual understanding. The DeepLab family of models [3], [4] further advanced the field by incorporating dilated convolutions and the atrous spatial pyramid pooling (ASPP) module, facilitating effective multiscale context aggregation. More recently, Transformer-based approaches [5], [6] have revolutionized semantic segmentation by utilizing self-attention mechanisms to capture long-range dependencies across spatial dimensions.

With the rapid advancement of Earth observation technologies, high-resolution remote sensing data from diverse sensors—such as optical, synthetic aperture radar (SAR), multispectral, and light detection and ranging (LiDAR) imagery—have become increasingly accessible. Among them, optical imagery and SAR imagery are the most widely employed data sources for LCC tasks. Benefiting from powerful backbone networks such as CNNs and Transformers in semantic segmentation, single-modality LCC methods have achieved remarkable success. For example, a scale-aware adaptation framework for remote sensing data was proposed in [7], enabling joint cross-location and cross-scale LCC. In addition, many CNN-based approaches leverage multiscale feature refinement and attention mechanisms to capture long-range dependencies and extract contextual representations [8], [9], [10], [11]. Despite their strong performance, these methods are inherently limited by their reliance on local receptive fields when modeling context. To address this, Transformer-based models have been introduced to remote sensing segmentation, effectively capturing global contextual information through self-attention mechanisms [12]. Furthermore, recent research has explored hybrid architectures that aim to integrate the strengths of both CNNs and Transformers [13], [14], [15].

Although deep learning-based segmentation methods for single modalities have demonstrated impressive performance, they are inherently limited in handling all real-world scenarios. This is primarily due to the inherent constraints of individual sensors, including limitations in wavelength range,

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Fig. 1. Example image of optical and SAR image pair. (Left) Optical image. (Right) SAR image.

operating conditions, and imaging modes, which often result in biased scene representations. Optical sensors, which rely on the passive detection of solar radiation reflected from the Earth's surface, can capture rich spectral and textural information. However, their performance degrades significantly under adverse conditions such as inclement weather or low-light environments. In contrast, SAR, an active sensing technology, is highly immune to climatic and lighting variations, enabling consistent data acquisition regardless of weather or time of day [16], [17]. Nonetheless, SAR images are typically affected by speckle noise and exhibit a relatively low signal-to-noise ratio. Fig. 1 presents an example of an optical–SAR image pair. Due to the inherent limitations and often incomplete information provided by a single modality, it is neither feasible nor reliable to generate a comprehensive, accurate, and consistent representation of the observed scene. Therefore, it is essential to integrate complementary information from optical and SAR modalities and suppress redundant features through multisensor data fusion. Despite its importance, only a limited number of studies have provided solutions for multimodal fusion LCC task [18], [19], [20]. For instance, a collaborative attention-based heterogeneous gated fusion network (CHGFNet) employs a collaborative attention-based heterogeneous gated fusion mechanism to achieve deep hierarchical fusion of optical and SAR features [21]. A dual-stream deep high-resolution network (DDHRNet) introduces a multimodal extrusion and excitation (MSE) strategy for feature-level fusion of SAR and optical branches [22]. A multimodal cross-attention network (MCANet) extracts high- and low-level features through a pseudo-Siamese architecture and designs three core modules to achieve multiscale feature fusion [23]. However, these multimodal LCC methods focus predominantly on deep feature fusion while neglecting the importance of structural consistency and mutual calibration. It is critical for effectively integrating heterogeneous modalities like optical and SAR imagery [24].

Optical imagery usually has rich texture and color properties, while SAR imagery appears in grayscale, reflects more ground structural details, and remains unaffected by illumination or adverse weather conditions. The purpose of fusing optical and SAR data for LCC is to leverage their complementary information to compensate for the individual limitations in image representation. Meanwhile, the structural consistency information of optical and SAR images can be used to establish effective correlations, which can fully improve the segmentation performance [25]. Recent studies focus too much on the learning of complementary features and the deep fusion of multimodal features while neglecting the significance of structural consistency and intermodal correction, which results in the loss or omission of associated information and degradation of classification performance.

To address the aforementioned challenges, we propose DCIFNet, a dual-stream correction-interaction-fusion multimodal network, designed to effectively correlate and fuse optical and SAR data. Specifically, this work first develops a differential feature correction and enhancement module (DF-CEM) that leverages bidirectional differential features to highlight modality discrepancies and correct each modality accordingly. These differential features emphasize both structural consistency and discrepancies between optical and SAR imagery of the same scene, thereby reducing the influence of noise. Furthermore, the parallel attention interaction module (PAIM) enhances cross-modal information exchange through dual-path attention mechanisms operating across channel and spatial dimensions. After that, the optical and SAR feature pairs are fused by the expert fusion module (EFM). A gating network dynamically assigns expert weights based on the input data characteristics, enabling adaptive selection and fusion of optical and SAR features. This mechanism effectively enhances segmentation performance for multimodal data.

The main contributions of the proposed methods are as follows.

1) A DF-CEM is proposed to refine both optical and SAR features. Specifically, DF-CEM exploits the differential information between modalities to guide cross-modality feature correction, which enables both streams to enhance complementary features and suppress modality-specific noise.
2) The PAIM is developed to achieve effective feature interaction. With parallel spatial and channel attention mechanisms, PAIM can not only focus on important spatial regions but also emphasize critical feature channels within multimodal features.
3) Inspired by the mixture of experts (MoE) mechanism, we design an EFM where optical and SAR features are regarded as separate experts. Equipped with the gate network, EFM adaptively filters redundant or irrelevant information and achieves a dynamic fusion of multimodal features.

II. Related Works

A. Optical or SAR Data LCC

Driven by advances in deep neural networks, a variety of effective single-modal LCC methods have been developed, including both CNN- and Transformer-based approaches. For instance, Alshehhi et al. [26] first proposed a CNN model that classifies features in high-resolution remote sensing images based on individual image patches, demonstrating the efficiency of CNN-based feature extraction for segmentation tasks. In [27], DeepLab was utilized to extract remote sensing features at multiple levels, effectively mitigating spatial resolution loss during decoding. DDCM-Net [28] introduced a dense

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dilated convolution merging network that incorporates both local and global contextual information for LCC. Owing to the powerful representation capacity of encoder–decoder architectures [29], numerous U-Net-inspired segmentation models have been developed for remote sensing applications. For example, a dilated stacked U-Net framework was proposed for RGB remote sensing imagery [30]. Furthermore, DeepUNet extended the U-Net architecture by introducing two additional types of skip connections, which improved segmentation performance on remote sensing data [31]. ResUNet-a [32] efficiently captured multiscale contextual features by integrating residual blocks, atrous convolutions, and pyramid pooling modules. More recently, an attention-based approach called MANet was proposed to address the underutilization of multiscale features in remote sensing segmentation tasks [11].

A variety of advanced methods leveraged multiscale and attention mechanisms to simulate long-range dependencies and enhance feature refinement, enabling the acquisition of finer spatial details and comprehensive global contextual information [8], [9], [33]. Furthermore, through the self-attention mechanism, many Transformer-based methods efficiently captured long-range dependencies in images [5], [6], [34]. For example, a global–local Transformer segmentation framework was proposed in [35] for semantic segmentation of remote sensing imagery, utilizing Transformer architectures in both encoder and decoder stages to ensure consistent feature representation. Owing to its superior performance in identifying water bodies and soil features [36], SAR imagery has become a critical data source in remote sensing segmentation. An attention-based FCN embedded with a multiscale attention module and a novel loss function was introduced in [37], achieving improved segmentation accuracy for SAR imagery. Due to the limitations of remote sensing satellite sensors, it is typically infeasible to simultaneously achieve high temporal, spatial, and spectral resolutions. Therefore, it is essential to integrate complementary information from multimodal data for LCC.

B. Optical and SAR Data Fusion LCC

Although deep learning-based LCC methods have advanced rapidly, most of them focus on single-modal data and therefore fail to address the inherent limitations of single-source remote sensing imagery. The synergistic use of multisource remote sensing data offers richer and more comprehensive information, thereby enhancing segmentation accuracy. Image fusion strategies can generally be categorized into three levels: 1) pixel-level fusion; 2) decision-level fusion; and 3) feature-level fusion [25]. Among these, most contemporary multimodal LCC methods primarily focus on feature-level fusion. Early implementations typically employed simple fusion operations, such as element-wise addition, multiplication, or channel concatenation. To address these limitations, Xu et al. [38] introduced a two-branch CNN framework that facilitates more effective feature fusion and achieved superior classification performance on multisource remote sensing data. Their work established that such dual-branch designs significantly outperform single branch in capturing complementary features. Extending this direction, an enhanced two-branch CNN was proposed in [39], which incorporated an adaptive feature fusion module based on a squeeze-and-excitation mechanism to improve the integration of multimodal data.

Optical imagery offers rich textures, colors, and radiometric features, while SAR imagery provides extensive structural information. In recent studies, various methods have been proposed to integrate these heterogeneous features with the goal of improving LCC performance. In [40], a two-branch pseudo-Siamese CNN combined with linear fusion methods was introduced to fuse optical and SAR features. A multimodal bilinear fusion network (MBFNet), along with the second-order attention-based channel selection module, was proposed in [24] to enhance the integration of fine-grained channel attention maps from optical and SAR modalities. Subsequently, Li et al. [21] proposed CHGFNet with collaborative attention and gated fusion modules and CEGFNet's end-to-end framework for hierarchical feature extraction [41]. In [42], optimized fusion strategies with cross-modal gating mechanisms have demonstrated superior performance in capturing complementary features through bidirectional information flow. Recent advances in multimodal remote sensing have yielded significant developments in both dataset construction and network architecture design. In DDHRNet [22], a multimodal LCC dataset was constructed using GF-2 optical imagery and GF-3 SAR imagery. Li et al. [23] subsequently contributed the large-scale WHU-OPT-SAR dataset, covering approximately 50 000 km^2 while proposing MCANet. It facilitates effective feature extraction, second-order hidden feature mining, and multiscale feature fusion for both low- and high-level representations.

In summary, while existing fusion methods demonstrate strong emphasis on complementary feature learning and deep multimodal feature integration, they exhibit two critical limitations: 1) insufficient consideration of structural consistency between multimodal data representations and 2) neglecting the potential of intermodal correction mechanisms. These overlooked aspects are critical for achieving robust and accurate integration.

III. Method

In this section, we first introduce the overall structure of the proposed DCIFNet and describe the Transformer involved. Afterward, three important modules in DCIFNet are introduced, namely, DF-CEM, PAIM, and EFM.

A. Network Structure

The overall architecture of the proposed DCIFNet is illustrated in Fig. 2. As an adaptation of the Transformer framework, DCIFNet adopts the encoder–decoder structure of SegFormer as its backbone. Specifically, DCIFNet constructs a dual-stream encoder consisting of two parallel branches to extract features from optical and SAR inputs. Each branch shares an identical architecture—a hierarchical Transformer encoder composed of four successive stages. These branches process optical and SAR data concurrently in a parallel yet interactive manner, with each stream tailored to capture modality-specific characteristics. To facilitate cross-modal calibration, DF-CEM is inserted between adjacent stages of the

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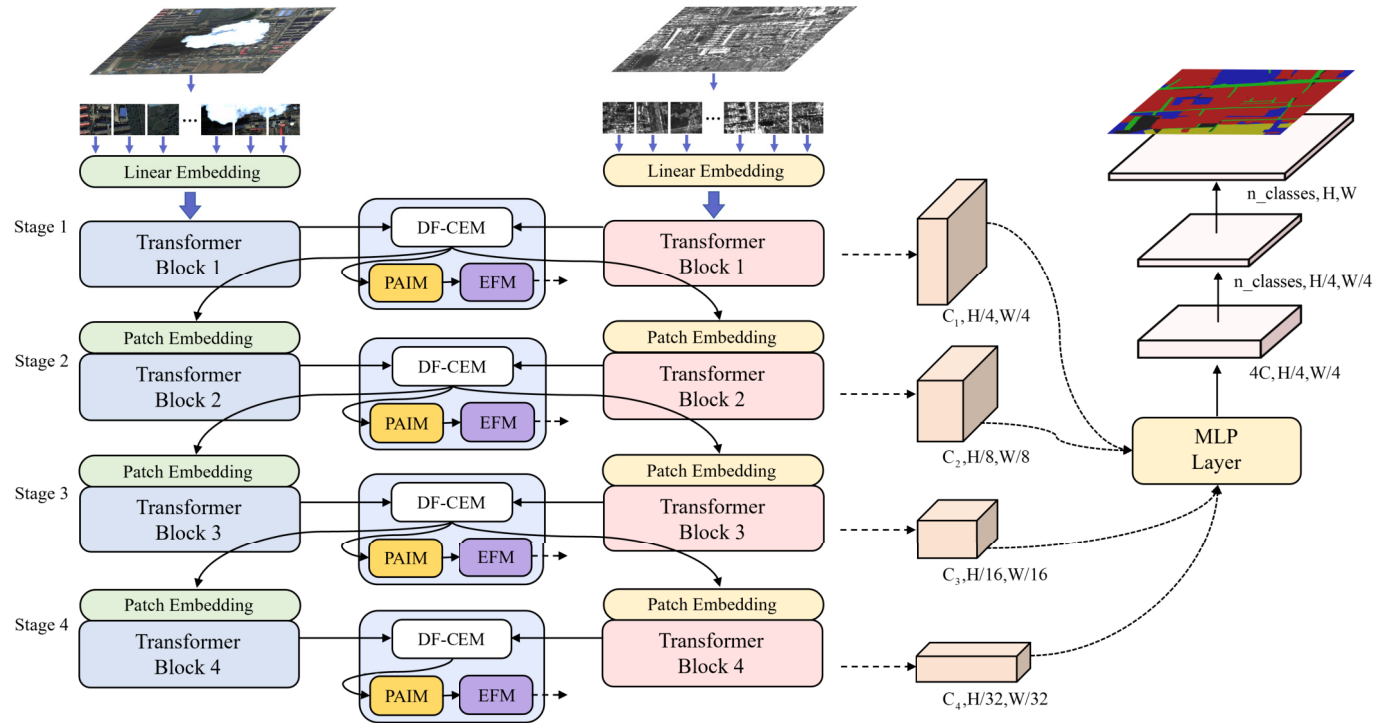


Fig. 2. Overall architecture of DCIFNet.

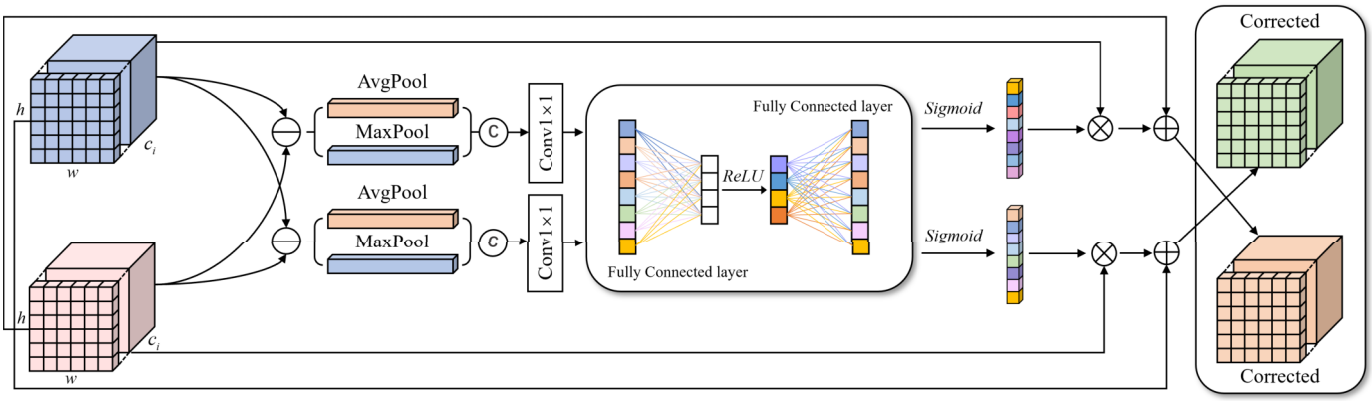


Fig. 3. Structure of the DF-CEM.

two backbones. It leverages differential features to rectify the optical and SAR representations, thereby transmitting enhanced features to the subsequent stages and progressively improving feature extraction quality. To further promote cross-modal interaction, PAIM is introduced to enhance feature exchange in both channel and spatial dimensions. Subsequently, EFM is employed to merge features at the same hierarchical level into a unified representation. The fused feature pairs from each stage are then passed through a multilayer perceptron (MLP)-based decoder to generate the class feature map. Finally, the output is upsampled to the original image resolution for pixel-wise LCC.

Given optical and SAR images $\mathbf{I}_{\text{opt}}, \mathbf{I}_{\text{sar}} \in \mathbb{R}^{H\times W\times 3}$, DCIFNet first applies an overlap patch embedding module to extract overlapping patches from each image using convolution operations and projects them into a feature space with dimension C_1. After the first stage of feature extraction, a pair of feature maps $(\mathbf{F}^1_{\text{opt}}, \mathbf{F}^1_{\text{sar}}) \in \mathbb{R}^{H/4\times W/4\times C_1}$ is generated by the two encoder branches. To capture multiscale features, a patch embedding module is utilized at the beginning of each stage. The output resolution at stage n is defined as follows:

$$\mathbf{F}^n_{\text{opt}} = \frac{H}{2^{n+1}} \times \frac{W}{2^{n+1}} \times 2^{n-1}C_1 \tag{1}$$

$$\mathbf{F}^n_{\text{sar}} = \frac{H}{2^{n+1}} \times \frac{W}{2^{n+1}} \times 2^{n-1}C_1 \tag{2}$$

where $n \in \{1, 2, 3, 4\}$ and H and W denote the height and width of the original image, respectively. Let $\{\mathbb{F}_{\text{opt}}, \mathbb{F}_{\text{sar}}\} = \{(\mathbf{F}^1_{\text{opt}}, \mathbf{F}^1_{\text{sar}}), \ldots, (\mathbf{F}^4_{\text{opt}}, \mathbf{F}^4_{\text{sar}})\}$ represent the multilevel optical and SAR feature pairs extracted from the four encoding stages. Each feature map pair $(\mathbf{F}_{\text{opt}}, \mathbf{F}_{\text{sar}})$ is first passed through DF-CEM, where differential correction is performed. The

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corrected features $\mathbf{X}_{\text{opt}}$ and $\mathbf{X}_{\text{sar}}$ are then forwarded to the next encoding stage for deeper and more refined representation learning. Simultaneously, the corrected feature pairs are also fed into PAIM and EFM to facilitate more effective cross-modal interaction and compact feature fusion. The output fusion feature of the nth EFM module is denoted as $\mathbf{X}_{\text{fused}}^{n} \in \mathbb{R}^{(H/2^{n+1})\times(W/2^{n+1})\times 2^{n-1}C_1}$. These multilevel fused features are subsequently passed to the MLP decoder to generate the class feature map $\mathbf{X}' \in \mathbb{R}^{H/4\times W/4\times n_\text{classes}}$, where $n_$classes denotes the number of target categories in the classification task. Finally, a bilinear interpolation upsampling layer is applied to $\mathbf{X}'$ to restore it to the original image resolution and produce the final prediction mask.

B. Differential Feature Rectification

As mentioned above, leveraging structural consistency between optical and SAR images can significantly enhance segmentation performance [25]. Nonetheless, effectively utilizing such structural information remains a key challenge that requires further exploration. We posit that differential computation provides an efficient means of revealing structural similarities and differences between the two modalities within the same scene, inherently capturing their contrasts in boundaries, shapes, and textures. To this end, we introduce DF-CEM, which exploits structural consistency information to progressively refine optical and SAR features at each stage of feature extraction, thereby mitigating the influence of noise. The effectiveness of this module is illustrated in Fig. 3.

Specifically, for the bi-modal features $\mathbf{F}_{\text{opt}}, \mathbf{F}_{\text{sar}} \in \mathbb{R}^{H\times W\times C}$ extracted at each stage, we define two differential features by element-wise subtraction

$$\mathbf{F}_{\text{os}} = \mathbf{F}_{\text{opt}} - \mathbf{F}_{\text{sar}} \tag{3}$$

$$\mathbf{F}_{\text{so}} = \mathbf{F}_{\text{sar}} - \mathbf{F}_{\text{opt}} \tag{4}$$

where $\mathbf{F}_{\text{os}} \in \mathbb{R}^{H\times W\times C}$ denotes the optical–SAR differential feature and $\mathbf{F}_{\text{so}} \in \mathbb{R}^{H\times W\times C}$ denotes the SAR-optical differential feature.

Subsequently, each differential feature map is then summarized through global average pooling and global max pooling along the spatial dimension, producing channel descriptors

$$\begin{aligned}\mathbf{C}_{\text{os}}^{\text{avg}} &= \mathcal{F}_{\text{avg}}(\mathbf{F}_{\text{os}}), \mathbf{C}_{\text{os}}^{\text{max}} = \mathcal{F}_{\text{max}}(\mathbf{F}_{\text{os}}) \\ \mathbf{C}_{\text{so}}^{\text{avg}} &= \mathcal{F}_{\text{avg}}(\mathbf{F}_{\text{so}}), \mathbf{C}_{\text{so}}^{\text{max}} = \mathcal{F}_{\text{max}}(\mathbf{F}_{\text{so}})\end{aligned} \tag{5}$$

where $\mathcal{F}_{\text{max}}$ and $\mathcal{F}_{\text{avg}}$ denote the global average pooling and global max pooling, respectively. Furthermore, $\mathcal{F}_{\text{max}}$ and $\mathcal{F}_{\text{avg}}$ can be defined as follows:

$$\begin{aligned}\mathcal{F}_{\text{avg}}(\mathbf{F}) &= \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbf{F}(i,j,:) \\ \mathcal{F}_{\text{max}}(\mathbf{F}) &= \max_{1\le i\le H, 1\le j\le W}\mathbf{F}(i,j,:).\end{aligned} \tag{6}$$

The average pooling captures the global context information by calculating the mean feature map intensity. On the other hand, the max pooling highlights the most prominent responses and focuses on high discriminative features. In order to jointly learn richer representations from differential maps ($\mathbf{F}_{\text{os}}$ and $\mathbf{F}_{\text{so}}$), the pooling descriptors derived from each differential feature are first concatenated. Subsequently, a 1×1 convolutional layer is applied to learn the weights that encapsulate differential features with rich structural information. This process can be described as follows:

$$\begin{aligned}\mathbf{Y}_{\text{os}} &= \text{Conv}_{1\times1}\left(\text{Concat}\left(\mathbf{C}_{\text{os}}^{\text{avg}} \parallel \mathbf{C}_{\text{os}}^{\text{max}}\right)\right) \\ \mathbf{Y}_{\text{so}} &= \text{Conv}_{1\times1}\left(\text{Concat}\left(\mathbf{C}_{\text{so}}^{\text{avg}} \parallel \mathbf{C}_{\text{so}}^{\text{max}}\right)\right).\end{aligned} \tag{7}$$

For the differential channel weights $\mathbf{Y}_{\text{os}}$ and $\mathbf{Y}_{\text{so}}$, an excitation-based block [43] is employed for learning. Each $\mathbf{Y} \in \mathbb{R}^{C}$ is passed through a two-layer MLP with reduction ratio $r = 4$ to capture nonlinear interchannel dependencies

$$\begin{aligned}\mathbf{W}_{\text{opt}} &= \sigma\left(\mathcal{F}_{\text{down}}\left(\text{RELU}\left(\mathcal{F}_{\text{up}}(\mathbf{Y}_{\text{os}})\right)\right)\right) \\ \mathbf{W}_{\text{sar}} &= \sigma\left(\mathcal{F}_{\text{down}}\left(\text{RELU}\left(\mathcal{F}_{\text{up}}(\mathbf{Y}_{\text{so}})\right)\right)\right)\end{aligned} \tag{8}$$

where $\sigma(\cdot)$ denotes the sigmoid function and $\mathcal{F}_{\text{down}}$ and $\mathcal{F}_{\text{up}}$ denote two MLP layers that perform dimensionality reduction and restoration of feature channels, respectively. Specifically, $\mathcal{F}_{\text{down}}$ projects high-dimensional features into a compressed latent space through $W_{\text{down}} \in \mathbb{R}^{(C/r)\times C}$, while $\mathcal{F}_{\text{up}}$ reconstructs the original channel dimensions to preserve information by $W_{\text{up}} \in \mathbb{R}^{C\times(C/r)}$.

Finally, the scaled differential channel weights are multiplied with the corresponding single-modal features to obtain the enhanced features, which are then added to the other single-modal features, resulting in the corrected bi-modal features. This process can be denoted as follows:

$$\begin{aligned}\mathbf{X}_{\text{sar}} &= \mathbf{F}_{\text{sar}} + \mathbf{W}_{\text{opt}} \circledcirc \mathbf{F}_{\text{opt}} \\ \mathbf{X}_{\text{opt}} &= \mathbf{F}_{\text{opt}} + \mathbf{W}_{\text{sar}} \circledcirc \mathbf{F}_{\text{sar}}\end{aligned} \tag{9}$$

where $\circledcirc$ denotes channel-wise pixel multiplication. Specifically, $\mathbf{F}_{\text{os}} = \mathbf{F}_{\text{opt}} - \mathbf{F}_{\text{sar}}$ reveals information present in the optical modality but absent in SAR, which helps generate $\mathbf{W}_{\text{opt}}$ for correcting SAR features accordingly. Conversely, $\mathbf{F}_{\text{so}} = \mathbf{F}_{\text{sar}} - \mathbf{F}_{\text{opt}}$ highlights geometric and structural information captured by SAR but missing in optical imagery, thus guiding the correction of optical features via $\mathbf{W}_{\text{sar}}$.

C. Parallel Attention Interaction Module

To address the respective limitations of optical and SAR imagery in feature representation, their fusion should fully exploit complementary information, which is crucial for improving classification performance. However, since features from different modalities reside in distinct feature spaces, traditional fusion strategies often result in information loss or distortion of critical features. We argue that introducing feature interaction into the fusion process can mitigate these issues. Therefore, we propose a novel PAIM, which facilitates feature exchange along both channel and spatial dimensions, thereby enhancing the learning of complementary representations from multimodal features. The architectural components of PAIM are illustrated in Fig. 4.

1) Channel-Wise Feature Interaction: Given stage n, the corrected optical feature $\mathbf{X}_{\text{opt}} \in \mathbb{R}^{h\times w\times c_1}$ and the SAR feature $\mathbf{X}_{\text{sar}} \in \mathbb{R}^{h\times w\times c_1}$ are first concatenated along the channel dimension, resulting in the $\mathbf{Z} \in \mathbb{R}^{h\times w\times 2c_1}$. Here, $c_1 = 2^{n-1}C$, $h = (H/(2^{n+1}))$, and $w = (W/(2^{n+1}))$. Since each channel of a

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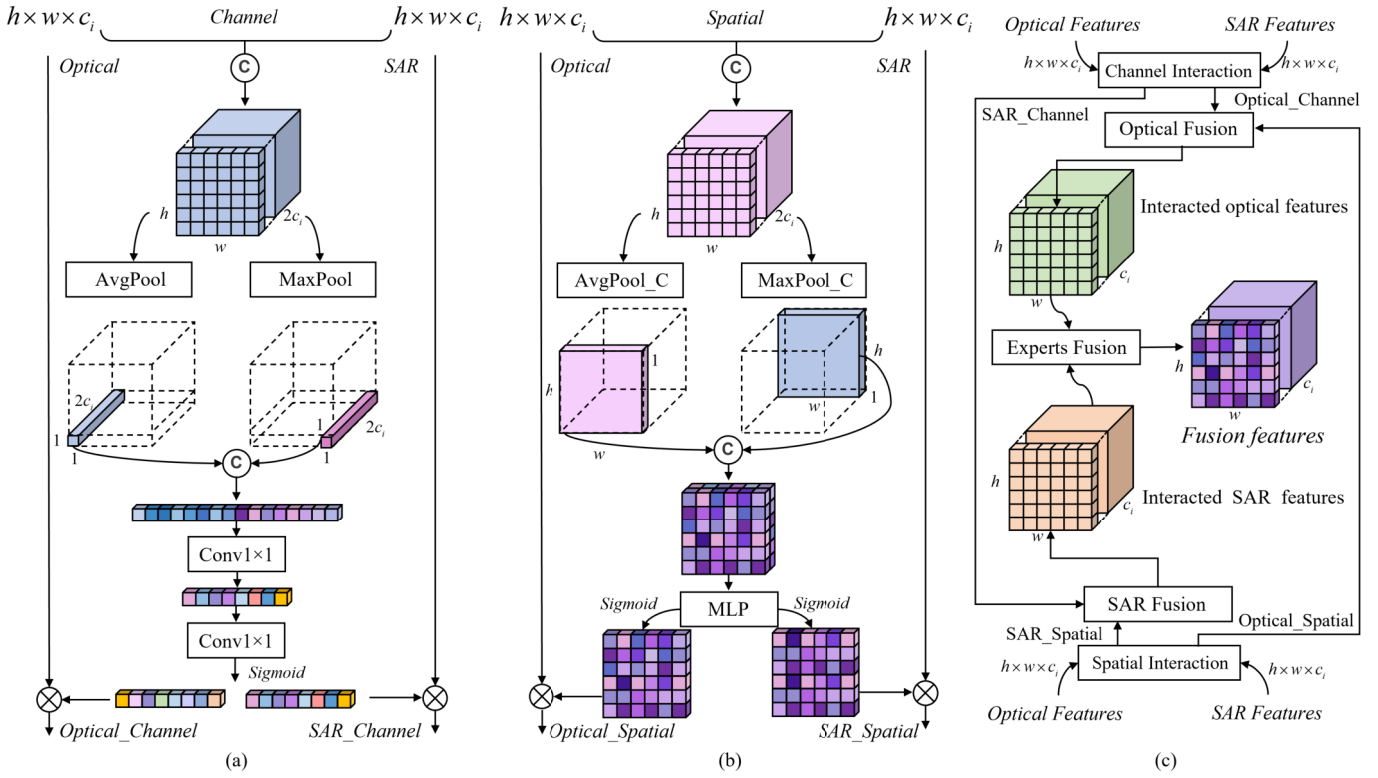


Fig. 4. Structure of the PAIM. (a) Channel-wise feature interaction module. (b) Spatial-wise feature interaction module. (c) Parallel attention interaction and fusion module.

feature map can be regarded as a feature detector, channel dependency primarily focuses on the relevant information within multimodal images. We employ two pooling strategies to obtain more comprehensive channel dependence. First, we apply average pooling and max pooling layers to calculate the statistical properties of the feature map on the channel. Then, $\mathbf{Z}_{\mathrm{A\&M}} \in \mathbb{R}^{1\times1\times4c_1}$ is obtained by concatenating the two to retain more information. After that, a 1×1 convolution layer with the rectified linear unit (ReLU) function is applied to learn the channel weights that incorporate global and local salient information. Afterward, we employ another 1×1 convolution layer to restore the dimension, followed by a sigmoid function to obtain the interaction-derived channel weights $\mathbf{W}^C$, which will be split into $\mathbf{W}^C_{\mathrm{opt}}$ and $\mathbf{W}^C_{\mathrm{sar}}$. This process is expressed by the following equation:

$$\begin{aligned} \mathbf{Z} &= \mathrm{Concat}\left(\mathbf{X}_{\mathrm{opt}} \,\|\, \mathbf{X}_{\mathrm{sar}}\right) \\ \mathbf{Z}_{\mathrm{A\&M}} &= \mathrm{Concat}\left(\mathcal{F}_{\max}(\mathbf{Z}) \,\|\, \mathcal{F}_{\mathrm{avg}}(\mathbf{Z})\right) \qquad (10) \\ \mathbf{W}^C &= \sigma\left(\mathrm{Conv}_{1\times1}\left(\mathrm{RELU}\left(\mathrm{Conv}_{1\times1}\left(\mathbf{Z}_{\mathrm{A\&M}}\right)\right)\right)\right) \\ \mathbf{W}^C_{\mathrm{opt}}, \mathbf{W}^C_{\mathrm{sar}} &= \mathcal{F}_{\mathrm{split}}\left(\mathbf{W}^C\right) \qquad (11) \end{aligned}$$

where $\mathcal{F}_{\max}$, $\mathcal{F}_{\mathrm{avg}}$, and $\sigma(\cdot)$ denote the global average pooling, global max pooling, and the sigmoid function, respectively. $\mathcal{F}_{\mathrm{split}}$ denotes the operation of splitting features along the channel dimension, producing channel weights $\mathbf{W}^C_{\mathrm{opt}} \in \mathbb{R}^{1\times1\times c_1}$ and $\mathbf{W}^C_{\mathrm{sar}} \in \mathbb{R}^{1\times1\times c_1}$. Subsequently, the derived channel weights $\mathbf{W}^C_{\mathrm{opt}}$ and $\mathbf{W}^C_{\mathrm{sar}}$ are applied to the original corrected optical and SAR features by channel-wise multiplication, which is formulated as

$$\begin{aligned} \mathbf{X}^C_{\mathrm{opt}} &= \mathbf{W}^C_{\mathrm{opt}} \circledcirc \mathbf{X}_{\mathrm{opt}} \\ \mathbf{X}^C_{\mathrm{sar}} &= \mathbf{W}^C_{\mathrm{sar}} \circledcirc \mathbf{X}_{\mathrm{sar}} \end{aligned} \qquad (12)$$

where $\circledcirc$ denotes channel-wise multiplication. $\mathbf{X}^C_{\mathrm{opt}}$ and $\mathbf{X}^C_{\mathrm{sar}}$ will be used for fusion in the EFM.

2) Spatial-Wise Feature Interaction: Simultaneously, we conduct multimodal feature interaction across the spatial dimension. Similar to the channel interaction, we first concatenate the corrected optical feature $\mathbf{X}_{\mathrm{opt}} \in \mathbb{R}^{h\times w\times c_1}$ and SAR feature $\mathbf{X}_{\mathrm{sar}} \in \mathbb{R}^{h\times w\times c_1}$ along the channel dimension to obtain $\mathbf{V} \in \mathbb{R}^{h\times w\times 2c_1}$. In order to identify the important regions within multimodal features, we apply weighting operations across the spatial dimension. In contrast to the channel interaction part, in the spatial interaction, average pooling and max pooling are performed along the channel dimension to generate spatial descriptors. Then, $\mathbf{V}_{\mathrm{A\&M}} \in \mathbb{R}^{h\times w\times 2}$ is obtained through concatenating the two feature maps. After that, an MLP layer is applied, which is composed of two 1×1 convolution layers assembled with the ReLU function. Finally, a sigmoid function is employed to acquire the interaction-derived spatial weights $\mathbf{W}^S \in \mathbb{R}^{h\times w\times 2}$, which are further split into two weight maps. The process can be denoted as follows:

$$\begin{aligned} \mathbf{V} &= \mathrm{Concat}\left(\mathbf{X}_{\mathrm{opt}} \,\|\, \mathbf{X}_{\mathrm{sar}}\right) \\ \mathbf{V}_{\mathrm{A\&M}} &= \mathrm{Concat}\left(\mathcal{F}_{\mathrm{max_c}}(\mathbf{V}) \,\|\, \mathcal{F}_{\mathrm{avg_c}}(\mathbf{V})\right) \qquad (13) \\ \mathbf{W}^S &= \sigma\left(\mathrm{Conv}_{1\times1}\left(\mathrm{RELU}\left(\mathrm{Conv}_{1\times1}\left(\mathbf{V}_{\mathrm{A\&M}}\right)\right)\right)\right) \\ \mathbf{W}^S_{\mathrm{opt}}, \mathbf{W}^S_{\mathrm{sar}} &= \mathcal{F}_{\mathrm{split}}\left(\mathbf{W}^S\right) \qquad (14) \end{aligned}$$

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where $\mathcal{F}_{\max_c}$ and $\mathcal{F}_{\text{avg_c}}$ denote the average pooling layer and the max pooling layer across the channel dimension, respectively, and $\mathcal{F}_{\text{split}}$ partitions $\mathbf{W}^S \in \mathbb{R}^{h\times w\times 2}$ along the channel dimension into $\mathbf{W}^S_{\text{opt}} \in \mathbb{R}^{h\times w\times 1}$ and $\mathbf{W}^S_{\text{sar}} \in \mathbb{R}^{h\times w\times 1}$. Furthermore, $\mathcal{F}_{\max_c}$ and $\mathcal{F}_{\text{avg_c}}$ can be defined as follows:

$$\mathcal{F}_{\text{avg}}(\mathbf{V})(i,j) = \frac{1}{C}\sum_{c=1}^{C}\mathbf{V}(i,j,c)$$
$$\mathcal{F}_{\max}(\mathbf{V})(i,j) = \max_{1\le c\le C}\mathbf{V}(i,j,c).$$

After that, the obtained spatial weights $\mathbf{W}^S_{\text{opt}}$ and $\mathbf{W}^S_{\text{sar}}$ are multiplied with the original corrected features along the spatial dimension, which are denoted as follows:

$$\mathbf{X}^S_{\text{opt}} = \mathbf{W}^S_{\text{opt}} \circ \mathbf{X}_{\text{opt}}$$
$$\mathbf{X}^S_{\text{sar}} = \mathbf{W}^S_{\text{sar}} \circ \mathbf{X}_{\text{sar}} \tag{15}$$

where $\circ$ denotes spatial-wise multiplication and $\mathbf{X}^S_{\text{opt}}$ and $\mathbf{X}^S_{\text{sar}}$ will serve as inputs for the subsequent module.

3) Feature Fusion: To derive the interacted optical and SAR features, we employ two EFMs to fuse the interaction-derived channel and spatial features $\mathbf{X}^C_{\text{opt}}$, $\mathbf{X}^C_{\text{sar}}$, $\mathbf{X}^S_{\text{opt}}$, and $\mathbf{X}^S_{\text{sar}}$, which are formulated as

$$\mathbf{X}^{\text{out}}_{\text{opt}} = \mathcal{F}_{\text{EFM}}\left(\mathbf{X}^C_{\text{opt}}, \mathbf{X}^S_{\text{opt}}\right) \tag{16}$$
$$\mathbf{X}^{\text{out}}_{\text{sar}} = \mathcal{F}_{\text{EFM}}\left(\mathbf{X}^C_{\text{sar}}, \mathbf{X}^S_{\text{sar}}\right) \tag{17}$$

where $\mathcal{F}_{\text{EFM}}$ denotes the EFM. After that, we also employ an EFM to perform the fusion of optical and SAR features

$$\mathbf{X}^{\text{out}}_{\text{fused}} = \mathcal{F}_{\text{EFM}}\left(\mathbf{X}^{\text{out}}_{\text{opt}}, \mathbf{X}^{\text{out}}_{\text{sar}}\right). \tag{18}$$

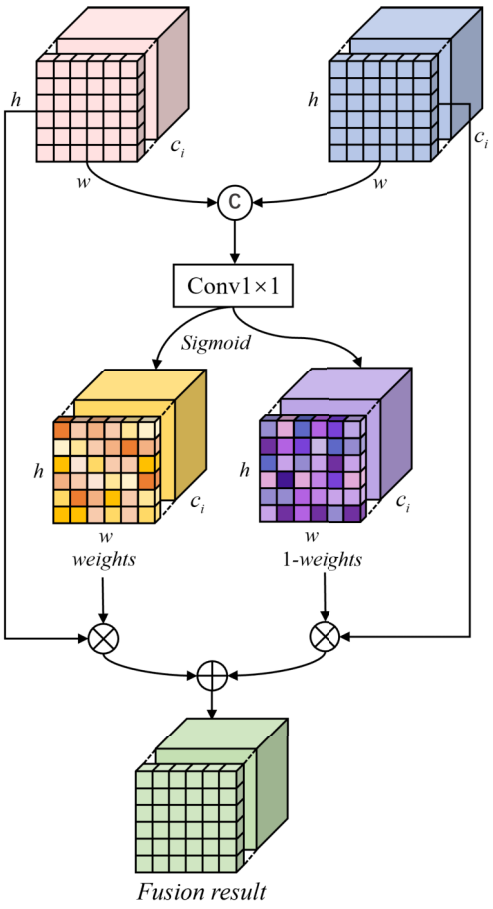


Fig. 5. Structure of the EFM.

D. Expert Fusion Module

The attention-based feature interaction module facilitates the effective learning of complementary information between optical and SAR features. However, conventional attention mechanisms and recent fusion strategies often struggle to filter out redundant information in real time. Inspired by the MoE paradigm, we design the EFM, which treats optical and SAR features as independent experts. The architectural components of EFM are illustrated in Fig. 5. Through a gating mechanism, EFM exhibits strong selectivity in filtering features, enabling robust fusion of informative content from different modality-specific experts. As shown in the rightmost module of Fig. 4, EFM is applied in two scenarios: one for fusing channel and spatial features and the other for fusing optical and SAR features.

To dynamically allocate expert weights for two input features $\mathbf{X}_1 \in \mathbb{R}^{h\times w\times c}$ and $\mathbf{X}_2 \in \mathbb{R}^{h\times w\times c}$, we introduce a convolutional gated network in the EFM

$$\mathbf{G}_1 = \sigma\left(\text{Conv}_{1\times 1}\left(\text{Concat}\left(\mathbf{X}_1 \,\|\, \mathbf{X}_2\right)\right)\right)$$
$$\mathbf{G}_2 = 1 - \mathbf{G}_1 \tag{19}$$

where $\mathbf{G}_1 \in \mathbb{R}^{h\times w\times c}$ and $\mathbf{G}_2 \in \mathbb{R}^{h\times w\times c}$ denote the expert weights of input features $\mathbf{X}_1$ and $\mathbf{X}_2$. The fused feature is then

$$\mathbf{X}_{\text{fused}} = \mathbf{G}_1 \odot \mathbf{X}_1 + \mathbf{G}_2 \odot \mathbf{X}_2. \tag{20}$$

1) Channel–Spatial Feature Fusion: As mentioned earlier, the interaction-derived channel and spatial features $\mathbf{X}^C_{\text{opt}}$, $\mathbf{X}^C_{\text{sar}}$, $\mathbf{X}^S_{\text{opt}}$, and $\mathbf{X}^S_{\text{sar}}$ are obtained by the PAIM. To derive the interacted optical and SAR features, we fuse their channel and spatial features by the EFM. Using optical features as an illustrative example, given $\mathbf{X}^C_{\text{opt}}$ and $\mathbf{X}^S_{\text{opt}}$, we first concatenate them along the channel dimension. Then, a 1×1 convolution layer is applied, followed by a sigmoid function to obtain the channel weight $\mathbf{G}^C_{\text{opt}} \in \mathbb{R}^{h\times w\times c_1}$ and spatial weight $\mathbf{G}^S_{\text{opt}} \in \mathbb{R}^{h\times w\times c_1}$. Afterward, the channel features and spatial features are multiplied with the channel weights and spatial weights, respectively, using the Hadamard product. Finally, the fusion optical features are obtained by adding the weighted channel features and spatial features. The fusion SAR features are generated following the same approach, as shown in the following equation:

$$\mathbf{G}^C_{\text{opt}} = \sigma\left(\text{Conv}_{1\times 1}\left(\text{Concat}\left(\mathbf{X}^C_{\text{opt}} \,\|\, \mathbf{X}^S_{\text{opt}}\right)\right)\right)$$
$$\mathbf{G}^S_{\text{opt}} = 1 - \mathbf{G}^C_{\text{opt}}$$
$$\mathbf{X}^{\text{out}}_{\text{opt}} = \mathbf{G}^C_{\text{opt}} \odot \mathbf{X}^C_{\text{opt}} + \mathbf{G}^S_{\text{opt}} \odot \mathbf{X}^S_{\text{opt}} \tag{21}$$
$$\mathbf{G}^C_{\text{sar}} = \sigma\left(\text{Conv}_{1\times 1}\left(\text{Concat}\left(\mathbf{X}^C_{\text{sar}} \,\|\, \mathbf{X}^S_{\text{sar}}\right)\right)\right)$$
$$\mathbf{G}^S_{\text{sar}} = 1 - \mathbf{G}^C_{\text{sar}}$$
$$\mathbf{X}^{\text{out}}_{\text{sar}} = \mathbf{G}^C_{\text{sar}} \odot \mathbf{X}^C_{\text{sar}} + \mathbf{G}^S_{\text{sar}} \odot \mathbf{X}^S_{\text{sar}} \tag{22}$$

where $\odot$ denotes element-wise multiplication.

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2) Optical–SAR Feature Fusion: For the optical and SAR features $\mathbf{X}_{\text{opt}}^{\text{out}}$ and $\mathbf{X}_{\text{sar}}^{\text{out}}$ obtained from channel and spatial interaction fusion in the previous stage, we similarly employ the EFM to perform feature fusion, which assigns weights to each modality and eliminates irrelevant components. The whole process of optical and SAR feature fusion is defined as

$$\begin{aligned}\mathbf{G}_{\text{opt}} &= \sigma\left(\text{Conv}_{1\times1}\left(\text{Concat}\left(\mathbf{X}_{\text{opt}}^{\text{out}} \parallel \mathbf{X}_{\text{sar}}^{\text{out}}\right)\right)\right)\\ \mathbf{G}_{\text{opt}} &= 1 - \mathbf{G}_{\text{sar}}\\ \mathbf{X}_{\text{fused}}^{\text{out}} &= \mathbf{G}_{\text{opt}} \odot \mathbf{X}_{\text{opt}} + \mathbf{G}_{\text{sar}} \odot \mathbf{X}_{\text{sar}}. \end{aligned} \tag{23}$$

Finally, the fusion features from each stage are sent to the lightweight decoder to further process.

IV. Experiments

In this section, we primarily describe the experimental results of the DCIFNet and other comparative methods on three multimodal LCC datasets from different perspectives. In all three datasets, the SAR and optical data correspond to the same geographic regions. Among them, two multimodal datasets contain cloudy images, while one dataset is cloud-free. The purpose of this comparison is to evaluate the generalization capability and robustness of the proposed method across various scenarios. The experimental results demonstrate that our method attains superior performance and effectively mitigates complex interference in remote sensing images.

A. Datasets

1) Xi'an Dataset: For the Xi'an region, Ren et al. [22] provided two groups of images: one with thick cloud coverage and the other with cloud-free. The cloudy subset is used to evaluate our method's performance in complex scenarios. The optical data (including RGB channels after preprocessing) were acquired from the GF-2 satellite, while the SAR data were collected in a slider spotlight mode with VV polarization. The Xi'an image has a resolution of 2304 × 3804. During training, the original image is cropped into patches of size 128 × 128, with a stride of 64, resulting in a total of 2122 images. The dataset is publicly available with a standardized 1:1 training–test split and contains five annotated land cover classes: building, road, farmland, water, and greenery.

2) Pohang Dataset: Likewise, in the Pohang region of South Korea, two groups of images are provided, and we focus on the cloudy group. In contrast to the Xi'an dataset, the clouds in the Pohang dataset are artificially added using the GNU Image Manipulation Program (GIMP) (as illustrated in Fig. 6), and they resemble fog or smoke. The size of the Pohang image is 10240 × 9216, and the image is cropped into 256 × 256 patches with a stride of 128. In total, 3087 images are used for training, and 3086 images are allocated for validation. The annotated categories are identical to those in the Xi'an dataset.

3) WHU-OPT-SAR Dataset: The WHU-OPT-SAR dataset is a large-scale multimodal land use classification dataset proposed in [23]. The dataset contains 100 optical and SAR image pairs of size 5556 × 3704, covering an area of 51448.56 km^2 in Hubei, China. The optical images originate from the GF-1 satellite, and the SAR images are collected by the GF-3 satellite, both with a resolution of 5 m. In addition, the original dataset is cropped into nonoverlapping patches of size 512 × 512. Furthermore, 7040 images are used for training, and 1760 images are allocated for validation. This dataset consists of seven classes: farmland, forest, city, village, water, road, and others.

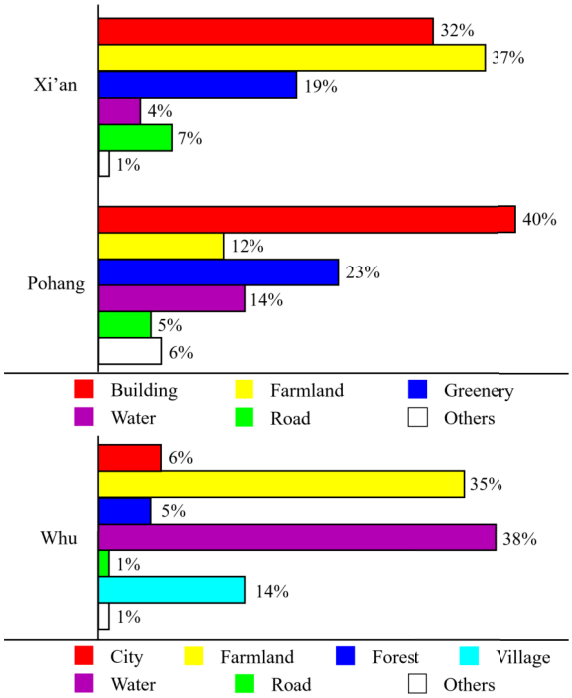


Fig. 6. Proportion of each semantic label in the Xi'an, Pohang, and Whu-OPT-SAR datasets.

Fig. 6 shows the proportion of each semantic label in the above three datasets. Meanwhile, we exclude the "others" category when performing quantitative evaluation on the Xi'an and Pohang datasets.

B. Implementation Details

1) Training Settings: Our network is built with the PaddlePaddle framework. We use the AdamW optimizer with a weight decay of 0.01 to train the model. In addition, we set the initial learning rate to 0.00006 and adopt the polynomial decay strategy. The other hyperparameters are set as follows: the number of epochs is 100 and the batch size is 4 for the Xi'an and Pohang datasets and 2 for the WHU-OPT-SAR dataset (due to potential out-of-memory errors, the batch size for the WHU-OPT-SAR dataset is set to 2). The loss function is cross entropy. All experiments are implemented on an NVIDIA GeForce RTX 3090 24-GB GPU. To ensure fairness, we do not load any pretrained models for all comparative algorithms.

2) Evaluation Metrics: In order to quantitatively evaluate the performance, overall accuracy (OA), mean intersection over union (mIoU), category intersection over union (category IoU), and category accuracy (category acc) are employed. These standard statistical metrics provide a fair basis for

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TABLE I
CONFIGURATION OF EXPERIMENTAL PARAMETERS FOR DIFFERENT MODELS

Model	Year	Optimizer	lr	Momentum/Beta	Decay
PSPNet [44]	2017	AdamW	0.001	(0.9, 0.999)	0.01
DeepLabV3+ [45]	2018	AdamW	0.001	(0.9, 0.999)	0.01
UNet3+ [46]	2020	Momentum	0.001	0.9	-
SegFormer-B5 [6]	2021	AdamW	6e-5	(0.9, 0.999)	0.01
SETR [5]	2021	SGD	0.001	0.9	0
DDRNet-23 [47]	2021	SGD	0.01	0.9	0.0005
SegNext [48]	2022	AdamW	0.001	(0.9, 0.999)	0.01
Segmenter [49]	2022	Momentum	0.001	0.9	-
SA-Gate [50]	2020	AdamW	0.001	(0.9, 0.999)	0.01
MCANet [23]	2022	Adam	0.001	(0.9, 0.999)	0.0001
CMX [51]	2023	AdamW	0.001	(0.9, 0.999)	0.01
ASANet [52]	2024	AdamW	0.0001	(0.9, 0.999)	0.05
FTransUNet [53]	2024	SGD	0.01	0.9	0.0005
DCIFNet (ours)	2025	AdamW	6e-5	(0.9, 0.999)	0.01

comparing the performance of the proposed method with other state-of-the-art methods. The OA and category IoU are based on the confusion matrix, which contains four items: true positive (TP), false positive (FP), true negative (TN), and false negative (FN). The OA is defined as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{24}$$

Each category IoU is defined as the ratio of the intersection of the predicted values and the ground truth to their union, which can be calculated as follows:

$$\text{category IoU} = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c} \tag{25}$$

$$\text{category acc} = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c} \tag{26}$$

where TP_c, FP_c, and FN_c denote true positives, false positives, and false negatives for the cth class, respectively. Furthermore, mIoU represents the average IoU across all categories.

C. Performance Comparison

We evaluate the proposed DCIFNet against 12 representative state-of-the-art methods, including PSPNet [44], DeepLabV3+ [45], UNet3+ [46], SegFormer [6], SETR [5], DDRNet [47], SegNext [48], Segmenter [49], ASANet [52], SA-Gate [50], MCANet [23], CMX [51], and FTransUNet [53]. The first eight networks are leading single-modal semantic segmentation models, which are employed to evaluate the benefits of multimodal fusion when optical and SAR images are concatenated as input. For these single-modal methods, we also conduct experiments using concatenated multimodal inputs to provide a fair comparison with our proposed approach. In contrast, the remaining five methods are specifically designed for multimodal data fusion, enabling a more direct comparison with DCIFNet. The hyperparameter settings used for different models are summarized in Table I.

1) Results on Xi'an Dataset: As presented in Tables II and III, the proposed DCIFNet achieves notable improvements in terms of OA, mIoU, category IoU, and category Acc when compared with existing methods. These results verify that DCIFNet effectively fuses multimodal data by extracting complementary and structurally consistent information from optical and SAR features, thereby generating robust and compact fusion representations. As shown in Table II, DCIFNet consistently outperforms state-of-the-art single-modal baselines across all land cover categories. Notably, compared with SegFormer-B5 using only optical data, our method achieves substantial improvements in IoU for the *Building* (7.12%), *Road* (10.1%), *Farmland* (5.26%), *Water* (5.18%), and *Greenery* (10.2%) classes after integrating SAR data. This demonstrates that the collaborative utilization of multimodal data can significantly enhance classification accuracy and mitigate the limitations of single-modal inputs.

Compared with other multimodal fusion methods, DCIFNet also exhibits superior performance in three categories: *Road*, *Farmland*, and *Water*. Specifically, on the *Road* class, DCIFNet provides a 5.16% improvement in IoU over SA-Gate. For the *Water* category, DCIFNet outperforms MCANet and CMX by 6.24% and 1.45%, respectively, indicating its enhanced ability to integrate complementary features derived from SAR data. Furthermore, although FTransUNet is designed for multimodal remote sensing data fusion, particularly for building extraction, DCIFNet achieves a 6.58% gain in IoU and a 2.67% improvement in accuracy on the *Building* class. These findings further validate the effectiveness of DCIFNet in exploiting both the complementary and structurally consistent characteristics of optical and SAR modalities.

Fig. 7 presents the qualitative segmentation results of five multimodal fusion methods listed in Table III. It is evident that our method produces the most accurate and consistent segmentation outcomes under both cloud-free and cloud-covered conditions. In groups (a)–(c), thick clouds and their shadows severely degrade the performance of competing methods, resulting in frequent misclassifications in occluded regions. In contrast, DCIFNet maintains notably higher accuracy in these challenging areas, demonstrating strong robustness to atmospheric interference. This advantage is largely attributed to the proposed DF-CEM, which effectively utilizes differential features between optical and SAR modalities to guide cross-modality correction. The results in (c) and (d) show that SA-Gate struggles to distinguish between the *Greenery* and *Farmland* classes, suggesting its insufficient exploitation of SAR-derived features. Although CMX and DCIFNet perform similarly in (c), (e), and (f), CMX produces several false predictions, highlighting its limitations in spatial precision and class discrimination. Conversely, DCIFNet demonstrates higher spatial coherence and semantic accuracy, particularly benefiting from the PAIM and EFM modules that enhance feature interaction and modality-aware fusion. Of particular note is the performance on the *Road* class, where DCIFNet exhibits better structural continuity compared to other methods.

2) Results on Pohang Dataset: As shown in Tables IV and V, DCIFNet achieves notable improvements in several

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TABLE II
USING SINGLE-MODAL DATA AS INPUT, THE PERFORMANCE OF DIFFERENT ALGORITHMS ON THE XI'AN DATASET

Method	Modal	Category IoU(%)					Metric	
		Building	Road	Farmland	Water	Greenery	mIoU(%)	OA(%)
PSPNet [44]	OPT	73.44	52.86	76.79	73.11	82.54	71.75	84.54
DeepLabV3+ [45]	OPT	86.88	65.69	91.52	80.89	82.10	81.42	92.12
UNet3+ [46]	OPT	84.45	57.53	87.23	79.20	78.50	77.38	90.18
SegFormer-B5 [6]	OPT	83.92	60.99	88.67	79.02	84.04	79.33	90.62
SETR [5]	OPT	80.67	39.90	85.90	67.29	77.04	70.16	86.51
DDRNet-23 [47]	OPT	84.84	65.42	86.34	81.04	73.84	78.30	90.60
SegNext [48]	OPT	84.40	49.47	86.22	78.32	81.98	76.08	89.03
Segmenter [49]	OPT	75.13	18.94	81.12	58.52	73.27	61.40	81.97
PSPNet [44]	SAR	85.41	52.27	83.99	64.93	86.18	74.56	87.94
DeepLabV3+ [45]	SAR	89.38	64.06	89.04	75.70	87.84	81.20	91.71
UNet3+ [46]	SAR	87.18	52.77	86.83	70.71	84.20	76.34	89.68
SegFormer-B5 [6]	SAR	89.43	63.87	88.31	75.33	88.59	81.11	91.60
SETR [5]	SAR	82.48	40.82	79.85	57.13	80.69	68.19	84.58
DDRNet-23 [47]	SAR	90.90	69.33	89.18	78.04	90.34	83.56	92.60
SegNext [48]	SAR	88.26	61.61	86.30	69.76	87.15	78.62	90.11
Segmenter [49]	SAR	72.41	11.73	73.63	43.51	73.81	55.02	77.11
DCIFNet (Mit-B5)	OPT-SAR	91.04	71.09	93.93	84.20	94.24	**86.90**	**94.34**

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS ON THE XI'AN DATASET USING OPT AND SAR MULTIMODAL DATA AS INPUT FOR LCC

Method	Category IoU (%)					Category Acc (%)					Metric	
	Building	Road	Farmland	*Water*	Greenery	Building	Road	Farmland	Water	Greenery	mIoU (%)	OA (%)
PSPNet [44]	79.54	63.12	89.53	72.01	92.04	94.98	88.61	92.00	76.59	95.82	79.25	89.25
DeepLabV3+ [45]	89.73	71.38	93.49	82.60	90.64	96.34	86.98	95.75	87.57	97.32	85.57	93.71
UNet3+ [46]	87.70	61.49	90.51	80.85	91.09	95.29	85.49	91.97	88.59	95.96	82.33	92.13
SegFormer-B5 [6]	89.28	68.38	91.18	80.07	90.16	94.32	86.93	94.42	88.71	93.80	83.81	92.76
DDRNet [47]	89.41	69.69	89.98	81.11	92.59	95.95	91.12	91.21	90.69	95.84	84.55	92.83
SegNext [48]	87.54	57.23	89.19	79.08	93.07	95.57	67.92	94.27	87.64	96.02	81.23	91.16
Segmenter [49]	79.49	34.03	83.00	62.56	87.81	86.77	62.20	89.09	77.61	94.65	69.38	84.90
SA-Gate [50]	88.65	65.93	88.97	80.61	86.43	95.72	89.22	88.95	90.34	97.82	82.12	92.12
ASANet [52]	88.89	60.53	92.48	82.51	94.51	95.07	73.04	96.25	89.78	96.71	83.79	92.96
MCANet [23]	88.13	63.21	92.09	80.18	94.02	95.86	82.14	95.07	85.48	96.45	83.53	92.50
CMX [51]	90.27	71.01	93.92	83.33	91.49	94.33	89.90	96.54	90.27	93.61	86.00	84.02
FTransUNet [53]	83.87	56.03	87.84	72.65	77.72	94.39	63.81	91.47	85.30	92.06	75.62	89.19
DCIFNet (Mit-B5)	91.04	71.09	93.93	84.20	94.24	93.84	90.65	96.69	91.72	96.01	**86.90**	**94.34**

challenging categories such as *Building*, *Road*, and *Farmland*, compared with both single-modal and multimodal baselines. Owing to the presence of thin clouds in this dataset, many regions remain distinguishable, allowing single-modal methods to achieve relatively high accuracy. As shown in Table IV, UNet3+ achieves an mIoU of 85.08% using optical data and an mIoU of 80.54% using SAR data. Benefiting from its well-designed fusion modules, DCIFNet achieves an mIoU of 87.00%, clearly outperforming other approaches. In particular, for the *Water* class, SegFormer-B5 attains 97.41% IoU using optical data and 96.32% with SAR. By incorporating SAR information, DCIFNet further improves this metric, demonstrating its superior fusion capability for extracting complementary features. In Table V, DCIFNet also surpasses several advanced multimodal segmentation methods. Notably, significant gains are observed in the *Building* category, outperforming FTransUNet by 3.4% in IoU, underscoring the advantages of our multilevel correction and fusion strategy.

Fig. 8 presents the visualization results of different methods on the Pohang dataset, which includes regions with thin cloud cover. Although thinner than in other datasets, these clouds still reduce the effectiveness of color and texture information in optical images, posing a considerable challenge for accurate segmentation. The visual comparisons show that DCIFNet achieves smoother object boundaries and significantly fewer false detections than competing methods. In particular,

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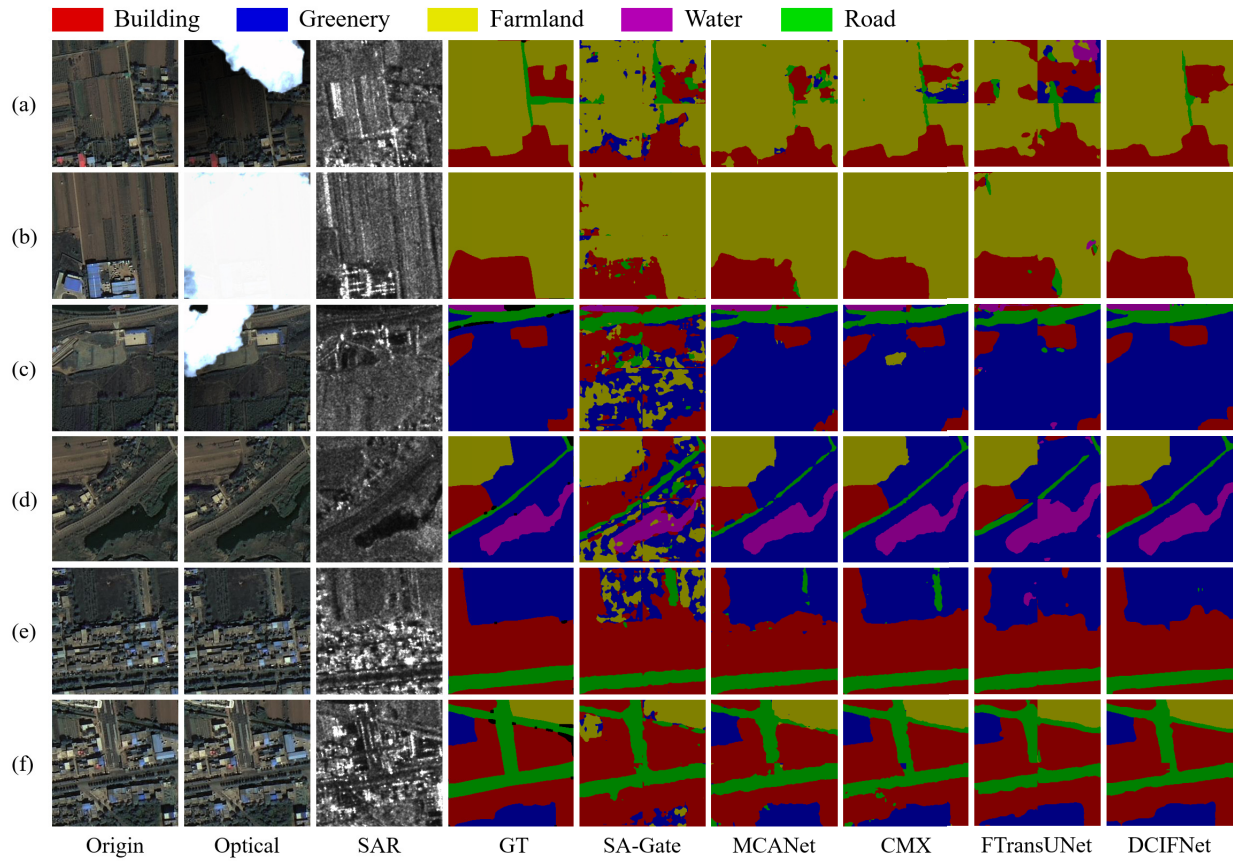


Fig. 7. Visualization of results on the Xi'an dataset. The first row shows the original optical image and the second row shows the cloud-covered optical image. The third row presents the corresponding SAR image and the fourth row displays the ground truth labels. SA-Gate, MCANet, CMX, FTransUNet, and DCIFNet are from left to right in each group.

TABLE IV

USING SINGLE-MODAL DATA AS INPUT, THE PERFORMANCE OF DIFFERENT ALGORITHMS ON THE POHANG DATASET

Method	Modal	Category IoU (%)					Metric	
		Building	Road	Farmland	Water	Greenery	mIoU (%)	OA (%)
PSPNet [44]	OPT	85.23	54.32	81.66	95.83	74.13	78.23	90.63
DeepLabV3+ [45]	OPT	87.95	53.82	81.48	96.65	77.49	79.48	91.47
UNet3+ [46]	OPT	91.12	65.58	86.75	97.73	84.23	85.08	94.01
SegFormer-B5 [6]	OPT	89.12	63.14	82.93	97.41	79.43	82.41	92.54
SETR [5]	OPT	79.31	31.13	69.64	95.04	67.92	68.61	85.52
DDRNet-23 [47]	OPT	90.52	70.03	84.97	96.17	81.51	84.64	93.48
SegNext [48]	OPT	90.46	64.72	85.15	97.13	82.25	83.94	93.42
Segmenter [49]	OPT	51.86	0.00	14.63	62.56	24.04	33.17	57.04
PSPNet [44]	SAR	90.22	59.77	83.53	96.08	78.01	81.52	92.51
DeepLabV3+ [45]	SAR	84.61	36.83	75.91	93.74	69.14	72.05	88.19
UNet3+ [46]	SAR	89.83	54.57	84.01	95.07	79.23	80.54	92.37
SegFormer-B5 [6]	SAR	90.52	62.45	84.01	96.32	78.97	82.46	92.85
SETR [5]	SAR	82.37	32.29	68.29	92.41	58.93	66.86	85.16
DDRNet-23 [47]	SAR	92.00	68.17	87.31	96.57	82.40	85.33	94.22
SegNext [48]	SAR	88.86	53.58	80.95	95.36	74.17	78.53	91.16
Segmenter [49]	SAR	66.26	0.00	39.56	58.39	0.00	32.84	64.82
DCIFNet (Mit-B5)	OPT-SAR	92.59	72.35	87.69	98.42	83.95	**87.00**	**94.76**

DF-CEM leverages bidirectional intermodal correction to mitigate the negative impact of thin clouds in optical imagery, thereby enhancing feature clarity and robustness. Meanwhile, PAIM effectively integrates structural cues from SAR data,

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TABLE V
LCC RESULTS OF DIFFERENT COMPARISON ALGORITHMS ON THE POHANG DATASET WITH OPT AND SAR MULTIMODAL DATA AS INPUT

Method	Category IoU (%)					Category Acc (%)					Metric	
	Building	Road	Farmland	Water	Greenery	Building	Road	Farmland	Water	Greenery	mIoU (%)	OA (%)
PSPNet [44]	89.96	64.85	87.19	91.95	82.33	95.08	87.96	93.01	93.58	88.39	83.26	93.21
DeepLabV3+ [45]	86.88	45.61	79.21	96.95	73.20	92.77	74.47	88.43	98.28	80.81	76.37	90.25
UNet3+ [46]	91.47	62.94	86.77	97.79	84.07	95.43	85.14	91.64	98.91	90.99	84.61	94.01
SegFormer-B5 [6]	88.75	52.81	82.11	97.68	78.41	93.11	79.26	89.65	98.80	87.98	79.95	91.91
DDRNet [47]	89.51	53.96	83.94	97.06	79.78	93.23	88.35	90.77	98.53	87.66	80.85	92.53
SegNext [48]	90.53	59.69	84.62	97.67	80.72	93.79	85.74	91.54	99.08	89.38	82.65	93.15
Segmenter [49]	80.94	27.93	70.82	95.93	68.96	86.61	62.69	82.95	98.71	82.96	68.92	86.36
ASANet [52]	81.97	17.98	72.57	96.80	69.22	94.99	18.94	81.76	97.67	83.90	67.71	86.89
SA-Gate [50]	84.73	23.43	74.31	95.88	68.29	87.38	88.67	92.35	99.69	73.05	69.33	87.96
MCANet [23]	88.66	46.88	82.45	97.51	78.65	92.63	67.19	92.35	98.94	87.21	78.83	91.62
CMX [51]	85.24	37.17	77.81	97.83	73.73	89.54	73.16	87.71	99.08	85.83	74.36	89.53
FTransUNet [53]	89.19	61.81	84.27	98.34	83.28	98.63	68.14	87.11	98.74	87.59	83.26	92.96
DCIFNet (Mit-B5)	92.59	72.35	87.69	98.42	83.95	95.81	91.02	92.43	99.57	91.28	**87.00**	**94.76**

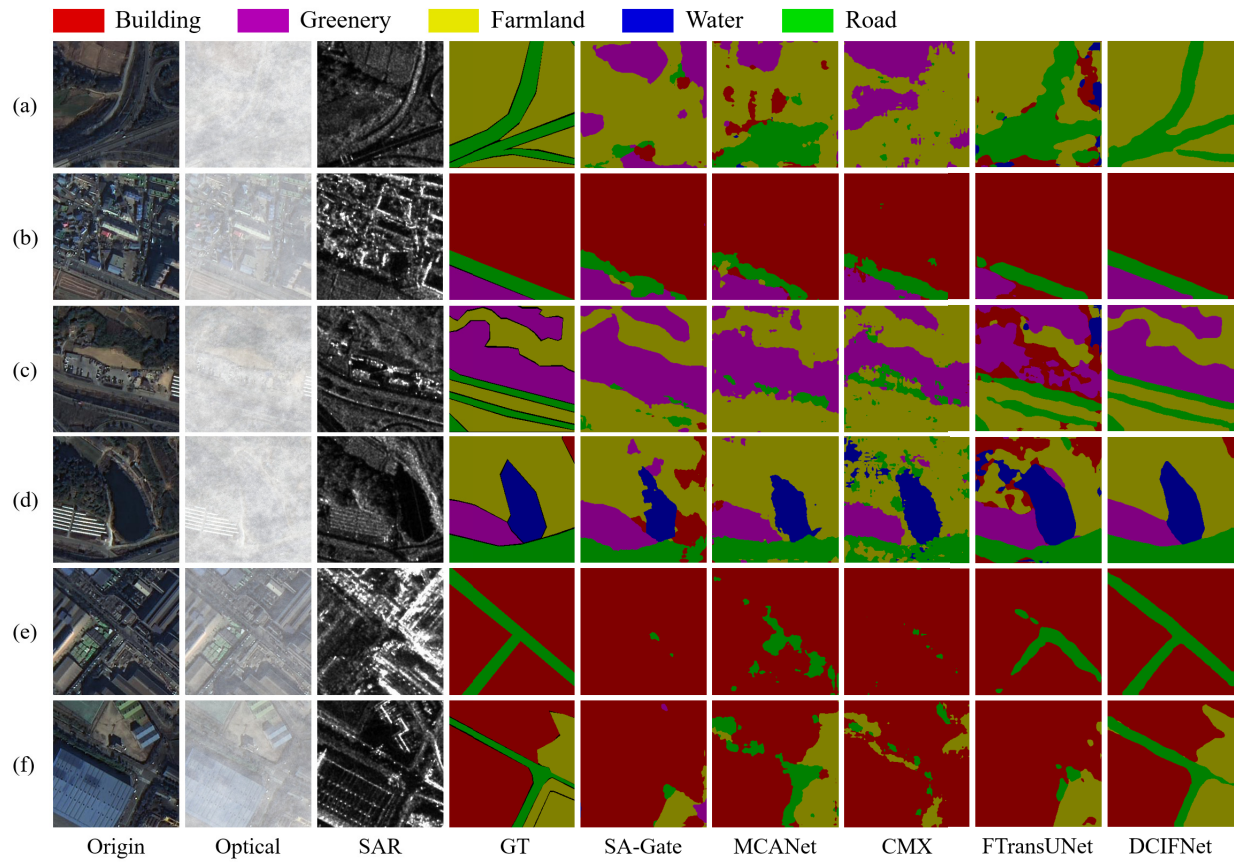


Fig. 8. Visualization of results on the Pohang dataset.

contributing to the preservation of fine boundary details in classes such as *Water* and *Road*, and facilitating accurate edge delineation. As seen in (a), (e), and (f), DCIFNet delivers precise road segmentation by fully leveraging SAR features, further validating its robustness under thin cloud conditions.

3) Results on WHU-OPT-SAR Dataset: The evaluation results of different models on the WHU-OPT-SAR dataset are summarized in Table VI. As a comprehensive, large-scale multimodal land use dataset, WHU-OPT-SAR encompasses complex remote sensing terrains and diverse scene types, thereby posing significant challenges for accurate

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TABLE VI
LCC RESULTS OF DIFFERENT COMPARISON ALGORITHMS ON THE WHU DATASET WITH OPT AND SAR MULTIMODAL DATA AS INPUT

Method	Category IoU (%)							Category Acc (%)							Metric	
	Farmland	City	Village	Water	Forest	Road	Others	Farmland	City	Village	Water	Forest	Road	Others	mIoU (%)	OA (%)
PSPNet [44]	62.55	56.47	42.56	49.84	70.16	32.37	23.58	76.83	74.71	63.50	58.36	85.48	59.27	58.58	48.22	75.27
DeepLabV3+ [45]	59.25	37.03	43.02	53.20	76.90	28.70	8.97	75.50	77.08	73.57	59.62	86.27	48.11	28.54	43.87	75.80
UNet3+ [46]	68.29	48.06	46.55	64.86	79.45	33.25	19.05	78.15	76.66	64.88	80.53	88.20	50.95	49.12	51.36	80.93
SegFormer-B5 [6]	67.85	52.57	47.55	65.70	79.06	33.63	18.27	79.50	69.94	66.69	80.44	87.48	54.12	45.26	52.09	80.97
DDRNet [47]	61.86	48.70	34.79	57.81	74.98	16.86	10.99	72.48	68.51	61.14	77.60	84.32	45.11	43.20	43.71	76.74
SegNext [48]	66.52	54.72	44.52	63.62	78.92	28.54	14.86	77.99	71.81	64.75	78.12	87.58	52.58	52.24	50.24	80.30
Segmenter [49]	62.13	48.24	33.22	58.32	76.95	9.37	6.66	73.88	64.56	55.29	76.83	85.51	29.24	30.28	42.13	77.01
ASANet [52]	64.95	49.63	43.71	58.16	81.76	25.40	13.16	77.16	74.78	61.37	71.73	92.26	32.31	14.98	48.11	81.52
SA-Gate [50]	56.97	23.69	33.97	50.07	72.52	10.37	0.34	72.86	73.35	54.74	58.01	82.40	43.90	18.10	35.42	72.68
MCANet [23]	66.30	53.91	46.33	62.36	78.70	23.19	5.50	76.84	72.52	68.65	79.67	86.46	53.39	46.89	48.04	80.13
CMX [51]	68.86	58.25	48.56	67.37	79.85	40.65	24.39	80.10	74.93	68.63	81.86	87.67	62.19	62.00	55.42	82.08
FTransUNet [53]	68.03	54.32	47.54	65.52	80.17	33.54	16.23	81.95	71.93	64.24	78.82	89.57	46.98	18.48	52.19	81.32
DCIFNet (Mit-B5)	69.12	59.14	49.09	67.65	80.08	39.88	26.17	80.12	74.93	66.37	81.47	89.08	61.03	56.88	**55.87**	**82.18**

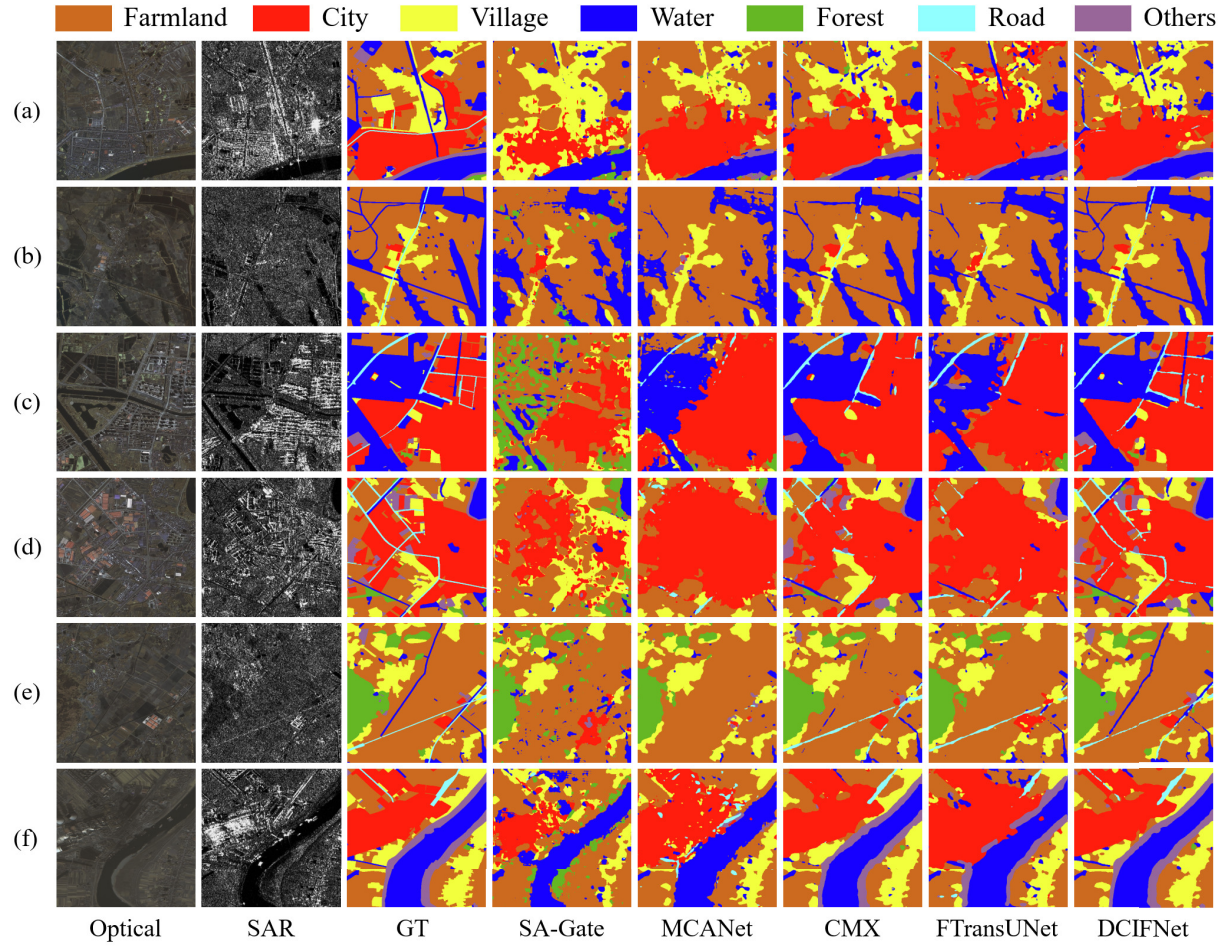


Fig. 9. Visualization of results on the WHU-OPT-SAR dataset.

classification. Despite these challenges, DCIFNet achieves the highest mIoU performance, surpassing SA-Gate, MCANet, CMX, and FTransUNet by 20.45%, 7.83%, 0.45%, and 3.68%, respectively, clearly demonstrating its superiority in segmentation accuracy. Moreover, DCIFNet consistently improves both mIoU and OA in the *City* and *Water* classes, attributed to its

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TABLE VII
ABLATION STUDY OF THE PROPOSED MODULES ON THE XI'AN DATASET

Modal	Modules				Category IoU (%)					Metric	
	EWS	DF-CEM	PAIM	EFM	Building	Road	Farmland	Water	Greenery	mIoU (%)	OA (%)
OPT					87.20	62.10	91.71	81.23	78.16	80.08	92.08
SAR					89.43	63.87	88.31	75.33	88.59	81.11	91.60
OPT+SAR	✓				90.04	68.86	92.69	82.02	91.30	84.98	93.54
OPT+SAR	✓	✓			90.06	69.71	93.32	83.85	92.17	85.82	93.89
OPT+SAR				✓	90.38	70.13	93.10	83.36	93.75	86.15	93.91
OPT+SAR			✓	✓	90.51	70.83	93.46	84.26	93.33	86.48	94.14
OPT+SAR		✓	✓	✓	91.04	71.09	93.93	84.20	94.24	**86.90**	**94.34**

TABLE VIII
ABLATION STUDY OF THE PROPOSED MODULES ON THE WHU-OPT-SAR DATASET

Modal	Modules				Category IoU (%)							Metric	
	EWS	DF-CEM	PAIM	EFM	Farmland	City	Village	Water	Forest	Road	Others	mIoU (%)	OA (%)
OPT					66.33	56.57	46.04	63.53	78.20	35.93	20.21	52.40	80.27
SAR					63.50	54.07	36.73	61.07	77.06	22.16	17.83	47.49	78.27
OPT+SAR	✓				67.49	57.32	46.95	64.76	79.12	37.68	20.38	53.38	81.02
OPT+SAR	✓	✓			69.05	55.92	49.25	67.09	79.83	38.04	24.34	54.79	81.91
OPT+SAR				✓	68.24	58.05	47.66	65.69	79.63	38.29	21.59	54.17	81.52
OPT+SAR			✓	✓	68.75	58.75	48.21	66.39	80.03	38.11	22.40	54.66	81.88
OPT+SAR		✓	✓	✓	69.12	59.14	49.09	67.65	80.08	39.88	26.17	**55.87**	**82.18**

effective exploitation of structural consistency between optical and SAR data, which enhances optical feature representation and gains higher accuracy.

Compared to other models, DCIFNet significantly reduces segmentation errors, with particularly notable improvements in the classification of *Road* and *Water* categories (as shown in Fig. 9). FTransUNet exhibits noticeable confusion between the *City* and *Village* classes in (a) and (d). In contrast, DCIFNet provides more accurate differentiation between these categories by effectively capturing their spatial boundaries. As shown in (b), (c), and (e), DCIFNet demonstrates higher segmentation accuracy and stronger spatial connectivity in delineating narrow structures such as roads and rivers. This reflects its ability to leverage both complementary features and structural cues from different modalities. Furthermore, in (e) and (f), DCIFNet achieves more precise classification for the *Forest* and *Farmland* classes, further highlighting its effectiveness in complex multimodal segmentation scenarios.

D. Ablation Study

Since we propose a new multimodal LCC network and introduce two important modules, it is highly valuable to investigate the individual contributions of each component to the overall performance. We perform ablation experiments on the Xi'an and WHU-OPT-SAR datasets. Tables VII and VIII present the results of the ablation experiments on different modules across the two datasets. The first two rows illustrate the performance of SegFormer-B5 with single-modal data inputs, providing a baseline for comparison with multimodal methods. The last four rows represent dual-stream SegFormer-B5 models, where different modules are applied at different stages for experimentation. The EWA module performs element-wise summation on the multimodal features generated at each stage.

1) Effect of Multimodal Data: As shown in Table VII, due to the interference of thick clouds, the prediction accuracy of optical images on the Xi'an dataset is lower than that of SAR images. Especially in building and road areas covered by thick clouds, the prediction IoU of SAR images is higher than that of optical images. We refer to the dual-stream Segformer-B5 with the EWS module (DSegB5-EWS). As shown in Table VII, both DSegB5-EWS (third row) and DCIFNet (sixth row) significantly improve IoU across all categories, with OA notably higher than that of single-modal baselines. These results demonstrate that the collaborative utilization of optical and SAR data maximizes the complementary information between the two modalities, enhancing segmentation performance. Compared with the DSegB5-EWS, DCIFNet yields a 1.92% increase for the mIoU and a 0.8% increase for the OA.

On the WHU-OPT-SAR dataset (shown in Table VIII), without cloud interference, the prediction accuracy of optical images is higher than SAR images. Furthermore, compared to using only optical data as input, DSegB5-EWS achieves only a 0.98% improvement in mIoU, which is significantly lower than the gain observed on the Xi'an dataset. This indicates that under normal environmental conditions, optical images alone can achieve relatively accurate classification. Nevertheless, our method still achieves a 3.47% increase in mIoU and a 1.91% gain in OA, demonstrating its ability to consistently extract

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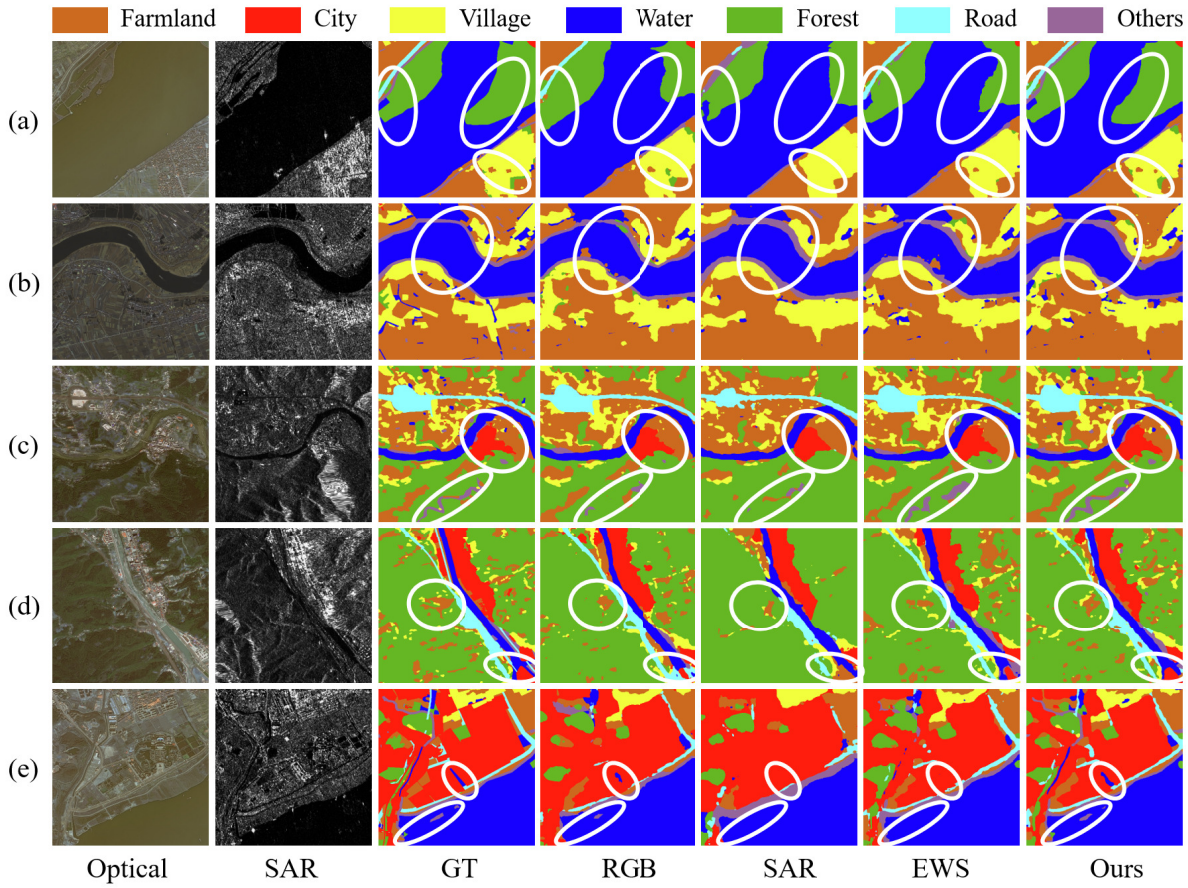


Fig. 10. Visualization of results on the WHU-OPT-SAR dataset. Optical images, SAR images, label images, OPT-Single results, SAR-Single results, DSegB5-EWS results, and DCIFNet results are from left to right.

and fuse complementary features across diverse environmental conditions.

To further demonstrate its effectiveness, Fig. 10 provides a visual comparison between DCIFNet, single-modal models, and DSegB5-EWS on the WHU-OPT-SAR dataset. These results highlight DCIFNet's ability to fully exploit cross-modal complementarity, leading to significantly improved LCC. The OPT-Single model captures rich texture and color information, performing well in farmland, city, and forest areas, but struggles in complex scenes where adjacent categories share similar spectral or textural features. The SAR-Single model, relying on structural features, performs well in delineating roads and water boundaries but struggles with farmland and village regions due to limited textural discrimination. As seen in Fig. 10, DSegB5-EWS tends to rely heavily on optical features when identifying large forest and water areas [see (a) and (b)]. In complex scenarios, the model becomes more susceptible to noise introduced by optical features. It is evidenced by the misclassification observed in (b), where water is incorrectly segmented as farmland. Conversely, in (e), the DSegB5-EWS favors SAR features when recognizing narrow roads and water, indicating difficulty in effectively integrating the strengths of two modalities. DCIFNet addresses these limitations through a more balanced fusion strategy, effectively mitigating the drawbacks of individual modality-specific weaknesses and producing superior performance.

2) Effect of the Proposed Modules: We select DSegB5-EWS as the baseline. The effectiveness of the proposed modules is validated by incrementally adding or replacing specific components.

Table VII reflects that after using DF-CEM on the DSegB5-EWS, the DSegB5-EWS-DF achieves improvements over the DSegB5-EWS in the road, water, and greenery categories on the Xi'an dataset, with IoU increasing by 0.85%, 1.63%, and 0.87%, respectively. On the WHU-OPT-SAR dataset, Table VIII also indicates that the DSegB5-EWS-DF achieves certain enhancements over the EWS model in the farmland, village, and water categories. However, these categories exhibit relatively poor performance in single-modal models. The results indicate that the proposed DF-CEM module effectively learns structural consistency information from both modalities, enabling the model to flexibly adapt to feature differences between modalities and improve segmentation performance across different categories. In addition, replacing the EWS module with the proposed EFM improves mIoU by 1.17% on the Xi'an dataset and 0.79% on the WHU-OPT-SAR dataset. This confirms that EFM's adaptive gating, which learns unique weights for the two experts, achieves a noticeably better fusion than pixel-level addition. Building on EFM, introducing the PAIM yields a further 0.33% gain on Xi'an and 0.49% on WHU-OPT-SAR. These increments demonstrate that PAIM effectively promotes cross-modal interaction and

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TABLE IX
ABLATION STUDY AT DIFFERENT STAGES

Stage				Metric		Category IoU (%)				
1	2	3	4	mIoU (%)	OA (%)	Building	Road	Farmland	Water	Greenery
				84.98	93.54	90.04	68.86	92.69	82.02	91.30
✓				85.63	93.83	90.34	69.49	93.10	83.43	91.81
	✓			86.32	94.22	90.80	70.85	94.09	83.78	92.09
		✓		86.50	94.20	90.91	69.99	93.32	84.77	93.54
			✓	85.81	94.01	90.55	69.09	93.39	84.17	91.83
✓	✓	✓	✓	**86.90**	**94.34**	**91.04**	**71.09**	**93.93**	**84.20**	**94.24**

captures additional complementary information. By integrating the DF-CEM module, DSegB5-EFM-PAIM-DF outperforms DSegB5-EFM-PAIM on the WHU-OPT-SAR dataset, achieving a 1.21% improvement in mIoU and a 0.3% increase in OA. The IoU metrics for all categories have also exhibited substantial enhancements.

However, as shown in Table VIII, DSegB5-EWS-DF shows a 1.4% reduction in IoU for the water category in comparison with DSegB5-EWS. The results indicate that element-wise summation is inadequate for capturing the intricate complementary relationships between two modalities and, in certain categories, may introduce additional noise, negatively impacting segmentation accuracy. In contrast, the proposed EFM and PAIM modules enhance the extraction of complementary information, enabling more effective feature selection and integration, thereby improving overall representation.

3) Effect of Different Stages: We adopt dual-stream SegFormer-B5 with the EWS module (DSegB5-EWS) as the baseline and conduct ablation experiments by inserting DF-CEM and PAIM at different encoder stages. As shown in Table IX, integrating these modules consistently improves both mIoU and OA across all stages. Notably, the second and third stages yield greater gains than the first and fourth. Upon further analysis, this phenomenon can be attributed to the nature of features extracted at different depths. Early stage features lack sufficient semantic information for reliable segmentation, while deep-stage features often contain limited spatial detail. The strong performance in *Farmland* categories at the second stage highlights the importance of local spatial, color, and texture features for multimodal fusion. Similarly, the superior results for *Greenery* at the third stage indicate these classes depend more on larger receptive fields and richer semantics. Therefore, applying DF-CEM and PAIM across all four stages yields the best overall performance, effectively meeting the high-precision requirements of diverse land cover categories.

V. CONCLUSION

In this article, a novel dual-stream multimodal correction-interaction-fusion framework, DCIFNet, is proposed for LCC in multimodal remote sensing data by fully utilizing complementary features and structural information from optical and SAR images. In particular, the differential feature correction and enhancement module uses the structural consistency information to refine optical and SAR features. Moreover, the PAIM is proposed to promote feature interaction along both channel and spatial dimensions, thereby enhancing the learning of complementary information within multimodal features. In addition, we propose the EFM as a selective feature fusion mechanism that effectively integrates the strengths of both modalities while suppressing noise effects, thereby enhancing feature representation. Numerical experiments on three datasets, Xi'an, Pohang, and WHU-OPT-SAR, have confirmed that the proposed DCIFNet can achieve superior performance under both cloud and cloud-free conditions.

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